

Macroeconomic determinants of high-tech migration in China: The case of Yangtze River Delta Urban Agglomeration

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ABSTRACT

With an increasing demand for high-skilled labors to fuel their high-technology industries, many Chinese cities have implemented a variety of incentive policies to compete for talents. As an attempt to evaluate the efficacy of local talent incentive policies and other factors that operate to attract and secure labors with needed skills and expertise, this study presents the first assessment of regional talent competition within the Yangtze River Delta Urban Agglomeration (YRDUA) on 150,000 sampled high-tech professionals, internalizing the interplay between the development of high-tech industries and economic growth, using a two-equation dynamic panel data model for the period from 2008 to 2018. Results show that while cities that are more ambitious in developing their high-tech industries appeal to a larger size of high-skilled talents, government interventions are more decisive. Also, the role of economic attractiveness in people's destination choices nowadays is not as important as before since people tend to leave for cities with better job opportunities and incentives for easier settlements. Policy incentives with lower-intensity rewards deserve more attention as it appears more effective in high-tech talent attraction. This study fills the research gaps in investigating macroeconomic determinants of high-skilled migration within one of the world's largest urban agglomerations.

1. Introduction

Technology and human capital have long been viewed as the driving forces of economic growth as talents often serve as “carriers” and contribute to technological progress through knowledge diffusion and innovation across organizations (Kerr & Lincoln, 2010). In recent years, rapid economic growth and industrial upgrade in China have led to a remarkable shift in the employment structure, which often requires a more skilled workforce, specifically in the high-tech industries. These talents, often well-educated and highly trained, can be mainly obtained through internal migration between two regions with clear differences in economic returns or amenities, e.g. living environments or opportunities for realizing personal aspirations (Lee & Kim, 2018). For local regions, implementing proper incentives can have a positive impact on the development of their high-tech industries. Among all, incentives towards talents play a vital role (Hong, Feng, Wu, & Wang, 2016). Therefore, in addition to higher public investment in educational resources, many Chinese local governments have been intensively introducing regulatory policies and incentives to attract high-skilled

migrants, aiming at boosting their human capitals.

The Yangtze River Delta Urban Agglomeration (YRDUA), one of the world's six largest mega-city regions and China's first world-class urban agglomeration (see Fig. A1), has benefited from the fast growth of overall productivity attributed to technological innovation and abundant human capital through talent production and attraction. Notably, as the core base of manufacturing in China, the YRDUA has a long history of developing heavy production entailing metals, automobile, electrical equipment and machinery (Feng, Chen, Song, & Cui, 2019). To cultivate new momentum for economic growth and industrial transformation, the YRDUA has paid great attention to innovation in key areas of high-end manufacturing industries, such as information technology, biopharmaceuticals, green energy vehicles, new materials, as well as modern services including finance, logistics, and technological research and development. Such industrial upgrades have spurred a higher demand for talents with technical and managerial skills.

Therefore, many cities in the YRDUA have implemented or are in the process of implementing aggressive policies to attract high-tech talents, aiming to meet their development targets in high-tech sectors.

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Driven by multiple factors, the high-tech talent¹ distribution landscape within the YRDUA has been changing. Fig. 1 clearly shows that in 2015, cities in Anhui Province, namely Ma'anshan and Chuzhou, enjoyed a net gain of high-tech talents. In 2018, despite sustaining a higher growth rate in the high-tech manufacturing industries, Ma'anshan suffered a great loss of high-tech talents. These talents moved to the northern Jiangsu Province and southern Zhejiang Province, where technological progress and talent competition remained moderate. It is not hard to see that the mobility pattern of high-skilled talents remains contested.

In reality, migratory decisions are usually affected by a variety of factors. Except that high-skilled migrants demand for a better job market, they are more likely to value a myriad of living-condition factors, such as education quality, provision of health infrastructure and services, and even the presence of a well-established professional network (Czaika & Parsons, 2017). The cases with high-skilled talents in China are even more complex as the internal migratory decisions are not only the results of regional attractiveness and individual preference, but also the complication of the *hukou* (household registration) system (Zhao, Liu, & Zhang, 2018).

Obviously, policy incentives come with a price. Therefore, local governments are in great need to understand the impacts of their talent policies on the actual selectivity of talents who are being pursued. In order to evaluate the efficacy of regional talent incentive policies and other factors that operate to attract and secure labors with needed skills and expertise in a rapidly developing mega-city region, this paper creates a unique and novel panel data and speaks to the latest spatial boundaries of the YRDUA (officially designated in May 2016). By factoring in the most recent local government incentives and the high-tech migration flow they might have subsequently brought, our research can provide decent remedies to the dated policy prescription. Most importantly, to better present the spatial and temporal changes in the migration process and deal with the endogeneity caused by the synergy of internal migration, innovative performance and economic growth, we apply simultaneous equation models with the System Generalized Method of Moments (sys-GMM) estimator.

This study, to our best knowledge, provides the first assessment of the gravity of new talent policies and unravels the dynamics of inter-city talent migration using the unique and up-to-date dataset of inward and outward high-skilled labor flows disaggregated by occupations and locations. The empirical results confirm the formation of radius patterns in the process of high-skilled migration as new “central cities” emerge after Shanghai within the YRDUA. We also evaluate cities' branding strategies in attracting high-skilled workers from a macroeconomic perspective. This paper adds to the literature of the traditional gravity model by considering simultaneously the latest policies for talent movements implemented by the destination cities, as well as social-economic factors at origin cities depending on migration history.

The rest of this paper is organized as follows. Section 2 provides additional background on China's internal migration and thereby proposes three hypotheses on the destination choices of high-tech talents. Data construction and research methodologies are introduced in Sections 3 and 4. Section 5 discusses the estimation results of all specifications and looks further into the evidence of the migration dynamics of high-tech talents. Section 6 links findings in this research to those in other countries and discusses the global use of city branding in attracting high-skilled migrants. We then conclude our empirical findings in Section 7.

¹ The China Statistics Catalogue of High-technology Industry Classifications defines high-tech manufacturing industries as medicine, aircrafts and spacecrafts, electronic equipment and communication equipment, computers and office equipment, medical equipment and measuring instruments, and electronic chemicals. We define high-tech talents as workers who are employed or job seekers who expect to get a job in the high-tech manufacturing sectors.

2. Literature review

2.1. Effects of high-skilled migration on regional economic development

A large body of literature has found extensive empirical evidence that high-skilled inter-regional migration can both positively and negatively affect the origin and destination regions in various ways. Observed from the production side of destination regions, new knowledge and skills are being carried to industries by high-skilled talents entering host cities (Zhou, Guo, & Liu, 2018). A higher concentration of skilled talents can enhance the productivity of the existing skilled workforce (Lee & Kim, 2018), and in turn create more skilled jobs and higher wage rates (Berry & Glaeser, 2005). In the long run, the skill endowment of the entire workforce can also be optimized with the mobility of high-skilled workers across different but related industries (Boschma, Eriksson, & Lindgren, 2014). In this way, the complementarity of high-skilled migrants in the existing workforce may bring a wider economic impact to the destination regions (Giannetti, 2003). From the consumption side, high-skilled migrants are more likely to improve the demographic structure because skilled workers are widely perceived with high comprehensive quality (Fang, Chan, & Lavelly, 2015). Besides, they may also raise the consumption level in the recipient regions because of a higher income (Nathan, 2015).

However, a consistent large-scale influx of migrants may also hamper the development of recipient regions over the long run. As population grows, the increasing demand for resources may overwhelm a city's infrastructure and service capacity (Bidandi, 2018), which consequently aggravates social problems such as higher crime rate or environmental degradation (Rafiq, Nielsen, & Smyth, 2017). This may lead to social instability and hinder the social integration of migrants (Bidandi, 2018).

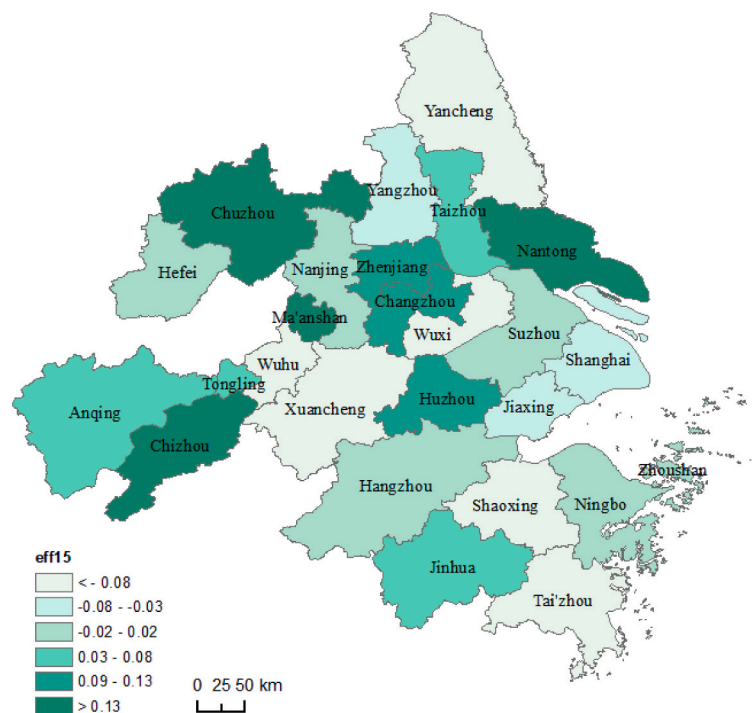
Different from the two-sided effects on destination regions, talent migration is mostly detrimental to the origin regions. The “brain drain” may worsen local talent shortage and slow down the accumulation of human capital, thereby undermining the overall productivity of origin regions (Ha, Yi, & Zhang, 2016). Chan and Wang (2008) discovered that internal migration originates from regional disparity that is in turn further exacerbated by an uneven geographical distribution of talents, as a vicious circle.

Putting all factors together, it is a challenging task to disentangle the effect of labor migration on regional development, and this is even more evident in the case of high-skilled inter-regional migration. Despite that migration may reinforce spatial inequalities and regional disparity (Zhou et al., 2018), most studies agree that high-skilled migrants play a positive role in improving the overall development of regions.

2.2. Macro-perspective determinants of high-skilled migration

The determinants of inter-regional migratory behaviors of high-skilled individuals have been extensively examined in the extant literature. Earlier studies derived from the *gravity model* mainly focused on destination population and distance variables (e.g. Ravenstein, 1976). Distance has long been recognized as a major impediment to literate migration (Schwartz, 1973). Nevertheless, DaVanzo (1978) pointed out that high-skilled individuals are more internally mobile and tend to move multiple times during their lives because they are more than capable to gather information and find suitable job opportunities in remote areas (Newbold, 1998). Recent research hinted that distance has gradually lost its explanatory power due to the nexus of the improved communication and transportation systems (Tamura & Masuda, 2017). In general, highly skilled workers mainly assess two regional aspects for their relocation decisions, namely labor market conditions and regional-specific amenities (Gonzalez-Leonardo & Lopez-Gay, 2019; You, Yang, & Fu, 2018), yet which one is the dominating factor is still an ongoing debate.

a. 2015



b. 2018

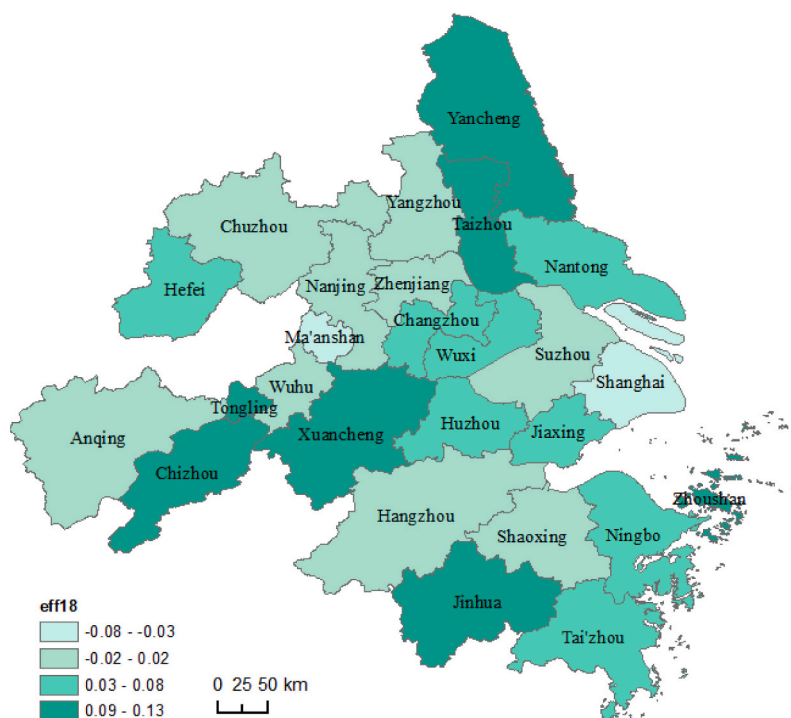


Fig. 1. Map of migration efficiency of high-tech talents in 2015 and 2018.

Note: Migration efficiency is calculated as the ratio of net migration (inflows minus outflows) to total in- and out-migration (Plane, 1984). Statistics of all cities during 2015 to 2018 can be seen in Table A1.

Source: Boss Zhipin.

One stream of literature asserts that regional imbalances in the local market force, which present itself by i.e., income (e.g. Yang, Cai, Qi, Liu, & Deng, 2017), employment opportunities (e.g. Wu, 2003), and foreign investment (e.g. Ng & Tuan, 2005) etc., have a decisive impact on the destination choices of skilled and educated individuals. Income

expectations and employment opportunities are both important concerns for inter-regional migrants. With no doubt, people tend to move to places where more employment opportunities are available and where job positions offer higher wages than they could have obtained in the origin regions after accounting for migration costs.

Although the study of economic factors has long been intriguing researchers from both theoretical and empirical attempts, it is only recently when attention is diverted to non-economic factors, such as climatic environment, cultural amenities and public services (Castro-Palaganas et al., 2017; Lee & Kim, 2018). These comfort-related factors seemingly carry more weights for high-skilled labor in their destination choices, as they are paying more attention to the standard of living and quality of life (Niedomysl & Hansen, 2010). Gibson and McKenzie (2011) found that professional elites prefer destination places with high levels of technological development, friendly environment, sufficient funding for research and good salaries. Also, urban growth is increasingly dependent on the cities' ability to provide services and consumer goods, which essentially attract high human capital (Glaeser, Kolko, & Saiz, 2001).

Similar to the findings in developed countries, the development of non-state-owned sectors (e.g. Cui, Nahm, & Tani, 2017; Wu, 2003), the living environment (e.g. Lee & Kim, 2018), the provision of public services (e.g. Liu, Qi, Cao, & Liu, 2015; Niu & Qi, 2015), and the institutional quality (e.g. Zhao et al., 2018) are gaining more importance in destination decisions in China. Cui et al. (2017) found that the development of non-state-owned sectors has been carrying an increasing weight in migrants' location choices. Chen (2011) stressed that the improvement of living conditions by eliminating unevenly distributed resources and opportunities will enhance a region's attractiveness. Cities that spend great efforts to provide a higher life quality and promote a more relaxed lifestyle and considerable green space will not only improve living conditions for existing residents, but also become more attractive to high-skilled individuals outside the regions (Yang et al., 2017; Zhou et al., 2018).

The household registration system in China, or *hukou*, is a major threshold of labor mobility, as it strongly tied to the access to local social welfare and public services. Although China has implemented a series of concrete reform policies to significantly lower or remove the registration threshold in lower-tier cities, the *hukou* system remains a strong gatekeeper for the attraction and retention of non-native household holders (Wu & Zhang, 2018). Shen (2013) and Zhao et al. (2018) both claimed that less restriction on the *hukou* system will exert both direct and indirect effects on the migration dynamics.

2.3. Literature gap and research hypothesis

While the increasing attention on high-skilled migration is gradually distinguishing itself from traditional framework, previous studies on high-skilled migration have several flaws that need to be addressed. To start with, studies assessing the effectiveness of migration policies in meeting its stated goals are sparse due to the complexity of recording policy elements and the contradictions of different regulatory measures in nature. Several researchers attempted to evaluate the immigration policies of a specific country in attracting high-skilled immigrants (Chacko, 2017; Ford, Jennings, & Somerville, 2015), but failed to address the poor quality of measurements in the mix of immigration regimes (Papademetriou & Sumption, 2013). Also, cities usually prefer to implement a series of regulatory measures with distinct stringency and time horizons, which also impeded the rapid progress of relevant research.

While the study of high-skilled migration has demanded for high-quality data, most empirical findings are based on aggregated datasets, e.g., questionnaire surveys or census data, and they are unable to capture the latest characteristics of high-skilled migrants either at the regional or at the occupational level. This might be the reason that few studies have emphasized on high-skilled talent migration due to insufficient statistics in this regard. Although Shen and Liu (2016) have noted skilled migrants in their research, they conceptualized them as those with tertiary education background. Obviously, it was an over-generalization of the characteristics of high-skilled workers. Since the higher education enrollment was expanded in 1998, China has

witnessed a surge of university graduates in the subsequent years (Levin & Xu, 2005), which in return enriched the meaning of being "high-skilled". Moreover, studies that utilize census data without disaggregating migrants by their occupations often fail to capture the nuance that distinguishes migratory behaviors of different groups. Lastly, the temporal changes in migration flows have largely been neglected in most existing studies.

Another challenge that analyses of internal migration from a macro perspective have to address is the endogeneity problem. Internal migration contributes to technological progress and spurs regional economic growth, which together in turn facilitate more migration. This process would consequently result in the endogeneity problem when all of them are simultaneously introduced to the same model. Specifically, Faggian, Modrego, and McCann (2019) noticed that the notion of endogeneity changes both subtly and fundamentally, as much of the regional growth stimulus is formed as the influx of externally acquired human capital. The endogenous growth that is specifically internal to destination regions is in the form of local knowledge spillover, resulting from interactions between the existing regional factors and the inflow of human capital (Zhou et al., 2018). Conversely, the aspect of endogenous growth which is specifically internal to the origin regions is the process of cumulative decline due to localized negative externalities (Boschma et al., 2014). By all means, human capital nowadays exhibits more as a dynamic flow rather than a static endowment. Therefore, the adoption of a static approach could lead to biases in both cognitive practice and modeling settings. As an empirical attempt to streamline possible evidence, the following three hypotheses are proposed and will be tested in this paper:

- H1.** High-tech talents are more likely to be attracted to places with higher levels of economic development.
- H2.** High-tech talents are more likely to be attracted to places with higher levels of high-tech manufacturing industry development.
- H3.** High-tech talents are more likely to be attracted to places with implementation of higher talent incentives.

To verify the three hypotheses and to fill the aforementioned literature gap, this paper constructed industry-specific talent inflows and outflows, allowing the determinants of high-skilled migration dynamics to be analyzed, and thereby moving beyond existing studies that examine data comprising of all skill levels. Notably, a two-stage least square technique is utilized to reduce the effects of reverse causality between high-tech manufacturing industry and talents, as well as economic growth and high-tech industry.

3. Data

To examine the migration determinants of high-tech talents, this study constructs a balanced panel, consisting of 26 cities in the YRDUA over the period from 2008 to 2018. In order to tackle the endogeneity problem as we have mentioned in the previous section, we further detail their relationship by disaggregating the complete database into two sub-datasets. One dataset is used to examine the relationship between the share of high-tech manufacturing sectors and per-capita GDP from 2008 to 2018, and the other one is used to test the efficacy of recent talent incentives except for some other macroeconomic variables that are ubiquitous throughout the literature such as economic growth, industry cluster, and amenities variables over the period from 2015 to 2018. The time span of the two equations are inconsistent mainly due to the limited availability of talent flow data that are accessed and assembled through a new channel (Teoh, 2018).² Details of the variable construction and data accessibility for the above-mentioned two

² Teoh defines data from online platform "new datasets" and argues the advantages of using "new datasets" over the traditional statistics.

specifications are described respectively in the following two stages.

3.1. Industry-related variables

In the first stage, we estimate how local economic development would affect local high-tech manufacturing industries. To distinguish advanced manufacturing industries from traditional ones, the share of high-tech manufacturing sectors in relation to the above-scale industries in terms of gross output value is used to measure the scale. Note that, while the compositions of high-tech manufacturing industries slightly differ among regions, they almost follow the same standard that was approved by the State Council.³ In this case, data of high-tech manufacturing industries in each region are harmonized to the greatest degree possible, therefore, no adjustment is made to the value. The gross output values of high-tech manufacturing industries of 26 cities from 2008 to 2018 are obtained from a variety of sources including the Science and Technology Department of Zhejiang,⁴ Jiangsu,⁵ and Anhui⁶ provinces, Shanghai Statistical Yearbook on Science and Technology, whereas data of 2018 are estimated by the annual growth rate recorded by the Provincial Statistical Yearbook and Reports on the Implementation of the 2018 Plan for National Economic and Social Development.

Many extant studies have investigated the impacts of macro-economic factors on the industry scale. To be specific, regional economic development level is predominantly represented by GDP per capita and it is deflated using the 2015 price. The role of government in advanced manufacturing sectors is commonly seen to be important. However, once it is empirically tested, ambiguous results can be observed. Some studies found that government grants may have exerted negative effects on industrial innovation (Hong et al., 2016). To include this factor in our study, the proportion of government expenditure on science and technology relative to the general public budget expenditure is used as an input. Comparably, foreign direct investments (FDI) (e.g. Ng & Tuan, 2005), and local financial resources (e.g. Guan & Yam, 2015) are also considered as critical driving forces for advanced manufacturing industries. Finally, on top of migration, local talent production is also attributed to regional development in China, which encourages us to use the university-based measure as a good proxy for talent attraction and production.⁷

3.2. Talent-related variables

In the second stage, we examine an array of factors that might impact migratory decisions. As an economically developed region, most cities in the YRUA have established decent infrastructure for high-tech manufacturing industries. The region has long been a reputable destination in China, which served as a strong foundation for a series of incentive policies to be carried out to further strengthen its attraction, especially since 2017. In this part of talent-related variables, we account for years from 2015 to 2018 to capture the characteristics of high-tech talent flows, mainly because this is the most intensive time window for policy-induced talent flow, and partially due to the availability of quality data that are retrieved from Boss Zhipin, one of the largest online recruiting platforms in China. By the end of 2018, it has recorded an accumulation of over 63 million online registers⁸ identified

as both employers and job seekers since its establishment in 2014. Changes in residence can be either reflected from a brief version of job seekers' full profile, recording the last company they worked for and the current position they are holding, or simply from their expectation for destination choices, which results in a recommendation list of jobs that specifically satisfy their expectations and has a higher chance of them migrating to the destination region. In doing so, short term trips such as traveling or daily commute are excluded, thereby better describing the effects of talent migration on a particular place over the long run.

To process the data, information of high-tech talents who prepare themselves for high-end manufacturing industry jobs is disaggregated by occupations and locations. The aggregation of monthly migratory flows into annual data, consistent with the construction method of other variables, appears to be a better representation of migration patterns since the trial-out talent flows⁹ have been smoothed out during this process. Only migrations that occurred between Chinese cities are included in this study. Additionally, the examination of talent inflows and outflows based on unilateral migration is constructed here to avoid a possible estimation bias induced by the simultaneous introduction of in- and out-migration within the same period to our models (Bauer & Zimmermann, 1999).

Among a variety of policy instruments for talent attraction,¹⁰ we focus on the housing subsidy and separate it into two categories according to its restrictiveness, one for exceptional talents endowed with higher human capital, and the other for younger talents with relatively lower human capital. To gauge the isolating effects of each policy element over time and across cities, we denote each category the value of one if the policy takes effect for a specific city-year. Details about both policy regimes and elements of 26 cities are included in Table A2.

Finally, a panel of non-economic indicators is also introduced here to ensure a well-specified model. In line with H2, the business environment of local technological development is proxied by the share of high-tech manufacturing industries, the estimates of which have been obtained from the first stage. In addition, myriad demographic and amenity variables that were remarked on in many previous studies also contribute to migratory decisions by individuals (Czaika & Parsons, 2017). For example, population density is calculated as population per square kilometers (km²). The provision of healthcare and education that high-skilled talents might consider as critical for their migratory decisions is measured by the number of local hospitals and schools respectively (Czaika & Parsons, 2017). The indicator of information and communications technology (ICT) coverage proxies to what extent a region is "technologically and culturally avant-garde" (Czaika & Parsons, 2017). All else equal, environmental conditions, acting as a "pushing" factors for migration (Hanna & Oliva, 2015; Hunter, 2005), is proxied by the ratio of greenspace in relation to the city land area, tacitly in the absence of measurements for regional environmental evaluation (Kloek, Buijs, Boersema, & Schouten, 2013).

All aforementioned variables collaboratively constitute a rich panel data which high-tech manufacturing industries share, followed by high-tech talent inflows, operating as dependent variables in two stages, respectively. Macroeconomic data are all obtained and assembled from the Statistical Yearbook, Statistical Communique of the National Economic and Social Development, the eleventh-, twelfth- and thirteenth Five-Year Plan for the National Economic and Social Progress in the level of provinces as well as cities over 10 years since 2008. Missing

³ Data source: <http://www.zjjxw.gov.cn>.

⁴ Data source: <http://www.zjkjt.gov.cn>.

⁵ Data source: <http://www.jssts.com>.

⁶ Data source: <http://www.ahkjt.gov.cn>.

⁷ In China, universities play an important role in facilitating talent mobility. Students usually have higher willingness and chances to work in the city where they receive higher education after graduation. Therefore, university-based measures are preferred over government expenditure on education considering that the latter is a summation of both higher education and basic education.

⁸ Records are as of December 2018.

⁹ Considering that many companies in China require a trial-out or probation period when an employee first joins the firm, inflows and outflows corresponding to the same talent within a three-month period can offset each other using our aggregation method.

¹⁰ Information is collected and assembled from the official websites of different local government units.

data in single years are constructed by using five-year moving average techniques. Details of data description are referred to Table A3, Figs. A2 and A3.

4. Econometric modeling

The unique panel datasets we have constructed are able to provide a better estimation with large number of samples across regions and over time. Nevertheless, estimation results are more likely to be challenged with several econometric issues. First, simultaneity bias is primarily introduced by the potential correlation of unobservable and time-invariant characteristics across examined regions and the explanatory variables. Disturbances are not likely to be constant across observations due to the heterogeneity of different cross-sections in this regard. This can be dealt with by applying the fixed effects model where the equation is first-differenced to remove the individual effects.

The second problem is the endogeneity and reverse causality between high-tech manufacturing industry and talents, as well as economic growth and high-tech industry. This problem hampers the unbiased and efficient estimates because the endogenous variables correlate with the current errors (Blundell & Bond, 1998). Besides, the presence of lagged dependent variables on the right-hand side notably in a dynamically determined system (Lee & Parasnis, 2014; Rogers & Wu, 2012) also possibly results in biased and inconsistent estimated coefficients. A possible solution is to use the two-stage least square technique only in the absence of heteroscedasticity and autocorrelation, which are ubiquitous in macroeconomic analysis of highly persistent and path-dependent variables, such as industrial output and institutional quality.

Returning to the facts in our study, the 26 cities constituting the YRDUA are mixed with developed and developing areas, allowing for the concurrence of similarities and variances. Taking culture as an example, it is of great possibilities but not easily measurable to hold a stable condition in a predictable period of time, thus, the unobservable regional differences embedded by the hidden effects on the right-hand side of the equations might hamper the efficiency and consistency of the estimation results. Meanwhile, the scale of either high-tech manufacturing industries or talent migration flows is perceived to be highly persistent and dynamically determined, and the lagged levels of dependent variables to the explanatory variable groups may also induce autocorrelation to the static modeling sets. Putting these all together, the use of static models would fail to provide either unbiased or consistent estimates.

To tackle these problems, we use a dynamic econometric approach based on the generalized method of moments (GMM), in which serial independence and homoscedasticity are well accounted for (Blundell & Bond, 1998). GMM was first used by Arellano and Bond (1991) to instrument the first-differencing endogenous variables with their lagged levels. Later, this technique was extended by Arellano and Bover (1995) and Blundell and Bond (1998), where equations were written in both difference and level forming. In doing so, the lagged first-differences are permitted as instruments in the level equations on the condition that the disturbances are independent and the difference series of dependent variables does not correlate with the remaining errors (Arellano & Bond, 1991). Therefore, the sys-GMM allows for an increase in efficiency and accuracy of estimation and better accounts for measurement errors, serial correlation, unobserved heterogeneity and endogeneity that is either oriented from *natural causes* or simply during the data generation process (Di Berardino, D'Ingiullo, Quaglion, & Sarra, 2018). Moreover, as empirically tested, the use of sys-GMM is proven to have provided a better outcome in the case of highly persistent and path-

dependent variables (Castello-Climent, 2008; Li, McHale, & Zhou, 2017).

With this in mind, our proposed models comprise two equations jointly estimated. Starting with the first specification examining the impacts of per-capita GDP and other factors on the share of high-tech manufacturing industries. The formulation is given by:

$$htshare_{i,t} = \alpha_1 htshare_{i,t-1} + \alpha_2 lnpcGDP_{i,t} + \alpha_3 X_{i,t} + \mu_i + \eta_t + \varepsilon_{i,t} \quad (1)$$

where $i = 1, 2, \dots, 26$ is the city index and $t = 2008, 2009, \dots, 2018$ is the time period covered by our dataset. $htshare_{i,t}$ is the share of high-tech manufacturing industries, and $lnpcGDP_{i,t}$ is the GDP per capita (logarithm transformed). The vector $X_{i,t}$ consists of a group of control variables of share of government expenditure on science and technology (in the percentage form), number of universities, inward FDI and financial savings (logarithm transformed), which are accordingly denoted as *govexp*, *edu*, *fdi*, and *finsave*. μ_i controls for time-invariant and city-specific individual effects, while η_t captures unobserved time effects. $\varepsilon_{i,t}$ is the residual disturbance.

Next, the specifications regressing talent inflows and outflows are separately formulated. Specifically,

$$\begin{aligned} lnTalentIn_{i,t} &= \beta_1 lnTalentIn_{i,t-1} + \beta_2 htshare_{i,t} + \beta_3 lnpcGDP_{i,t} + \beta_4 lnpopden_{i,t} + \beta_5 \\ &X_{i,t} + \mu_i + \eta_t + w_{i,t} \end{aligned} \quad (2)$$

$$\begin{aligned} lnTalentOut_{i,t} &= \beta_1 lnTalentOut_{i,t-1} + \beta_2 htshare_{i,t} + \beta_3 lnpcGDP_{i,t} + \beta_4 lnpopden_{i,t} + \beta_5 \\ &Z_{i,t} + \mu_i + \eta_t + v_{i,t} \end{aligned} \quad (3)$$

where $i = 1, 2, \dots, 26$ is the city index and $t = 2015, 2016, 2017, 2018$ is the time period tracking talent flows. The talent data are first retrieved from Boss Zhipin and then summarized into two distinct variables of *TalentIn* and *TalentOut* to represent high-tech talent inflows and outflows, respectively. Population density is denoted as *popden*. The amenity vector $X_{i,t}$ is consisted of four variables, namely the number of hospitals (*hospnum*) and universities (*univnum*), the coverage of ICT (*ictcov*), and green space (*greens*). The two-component error terms are comparable with those in Eq. (1). Besides, GDP per capita, the share of high-tech manufacturing industries, and policy variables are treated as endogenous variables, and instrumented with their second- or higher-order lags. Regarding the mechanism of sys-GMM estimator, employing too many instruments might reduce the accuracy of confidence intervals. Therefore, SARGAN test and Arellano-Bond test are conducted for over-identifying of instruments and autocorrelation.

5. Results

Our analysis proceeds in two conceptual steps: First, we report the results of the first specification estimating the relationship between the two variables of particular interest, i.e., high-tech manufacturing industries and GDP per capita; Then, we look into the evidence testing the efficacy of talent incentives except for other demographic and socioeconomic factors that often matter for talent attraction and retention.

The estimation results related to the first specification using sys-GMM estimator, OLS, and fixed effects are listed in Table 1. Model (1) in the first column reports the baseline regression results that include estimates of the lagged dependent variable, GDP per capita, government expenditure, and foreign investments. Models (2) and (3) add measures of local education capability and financial resources. To clarify the difference with static models, estimates of OLS specification

Table 1
Estimation results of specifications regressing high-tech share.

Independent variables	Model (1)	Model (2)	Model (3)	OLS	Fixed effects (two-component error)
L. htshare	0.662*** (0.065)	0.647*** (0.105)	0.611*** (0.078)		
D. htshare	0.268** (0.105)	0.293*** (0.107)	0.310*** (0.087)		
lnpcgdp	−0.020 (0.023)	−0.015 (0.019)	0.008 (0.062)	0.146*** (0.038)	0.087* (0.043)
govexp	−0.182 (0.436)	−0.335 (0.748)	−0.375 (0.528)	0.768 (0.940)	0.234 (0.575)
lnfdi	0.010* (0.006)	0.015* (0.009)	0.026** (0.011)	0.019 (0.016)	0.045*** (0.010)
edu		−0.001 (0.001)	−0.001 (0.002)	0.000 (0.002)	−0.006*** (0.002)
lnfinsave			−0.017 (0.032)	−0.040 (0.024)	−0.003 (0.022)
Number of instruments	33	34	35		
SARGAN <i>p</i> -value	0.5369	0.6988	0.7328		
AR(2)	0.4688	0.2705	0.4589		
Number of observations	208	208	208	260	260

Note: *, **, *** denote statistically significant at 10%, 5% and 1% level. L. htshare and D. htshare denote the first and second lags of dependent variable. Robust standard errors are in parentheses. City and year dummies are included in all models. The lagged dependent variables, per capita GDP, inward FDI, financial savings are treated as endogenous variables; Second- and higher-order lags of the levels are used to instrument endogenous variables. No external instrumental variables are used.

as well as with the fixed effects model are also reported in the last two columns. All models include origin-time fixed effects to guarantee the best possible comparison of different models.

The validity of using instruments to identify these dummies in a dynamic sys-GMM estimation is supported by the SARGAN test of overidentifying restrictions. The appropriateness of estimation tools is also guaranteed by the Arellano-Bond tests for serial correlation of error terms. Furthermore, the *p*-value of SARGAN test is found to be robust, which implies that the null hypothesis that all instruments used are valid and cannot be rejected. The reported AR(2) tests indicate that we cannot reject the null hypothesis that no significant autocorrelation occurs in error terms. The two-year lagged dependent variables are statistically significant at the 1% level, which also verifies the use of dynamic models. Comparably, OLS estimates are not robust to dynamic panel estimation in this case. It can be explained by Di Bernardino et al. (2018) that when applying static models to highly persistent variables, especially from an economic sense, chances are slim for making rapid adjustments to any shocks occurring within the same period of time.

As shown in Models (1)–(3), the impacts of per capita GDP are insignificant and not a clear-cut. This corroborates with the fact that regions within the YRDUA are both fast paced and sometimes interleaved with respect to their development level. Take Shanghai for example, as one of the most developed cities in China, a sharp decline in manufacturing industries has been witnessed during the last five years. It seems that the manufacturing industry has lost its chances in Shanghai until local authorities officially announced a 25% target in the beginning of the 13th Five-Year Plan in 2016, in order to be well prepared for the post-industrial era when support and modern services are largely demanded. Conversely, Anhui Province, as the only developing

province in the YRDUA, has spared substantial efforts and caught up with the fast development of the high-tech industries.

When looking at the government efforts, we find it interesting that government expenditure on science and technology is not statistically significant and even imposes negative effects on the development of advanced manufacturing industries. This is somewhat surprising at first glance; however, similar findings were also found in Guan and Yam (2015) and Hong et al. (2016). One possible reason is that a large proportion of government spending is used for construction and maintenance of project-related equipment and infrastructure. Besides, the crowding-out effects of government grants on private funding, along with the lack of punishment mechanism of the inappropriate use of grants, could possibly hamper the technological improvements.

Furthermore, neither local educational nor financial resources have a positive association with the advanced manufacturing industries, and the relationship is rather weak. Although the coefficients of these two variables are not statistically significant at conventional levels, their economic significance, such as venture capital and human capital, weigh in on the expansion of high-end manufacturing and support service industries (Duan, 2008).

In contrast to the weak relationship between domestic efforts and high-tech industry development, the coefficients on inbound FDI become larger in magnitude and attain a higher level of significance when educational and financial resources are added into the specification. This result collaborates with that of Hong (2009), proving that the utilization of foreign investments has indeed encouraged a faster progress in the advanced equipment manufacturing pipeline.

Furthermore, to identify the isolating effects on the migratory and emigratory decisions, Table 2 lists the results regarding high-tech talent inflows and outflows, respectively. Model (1) in Table 2, specified as the basic model, concentrates on the demographic and economic factors. It is then augmented by policy variables arrayed in isolation and combination in Models (2)–(4). Models (5)–(8) extend the basic model by including four additional quality-of-life variables.

Either high-tech talent inflows or outflows, to some extent, are impacted by the immigratory and emigratory decisions of the previous year, which agrees with the use of the dynamic model. The *p*-value of SARGAN tests of all models fails to reject the null hypothesis that all instruments are valid, except for the baseline models without amenities variables. The results of Arellano-Bond test (AR(2)) fail to reject the null hypothesis that no serial correlation exists in the current error terms.

We find a countervailing effect in the economic and demographic conditions which are represented by GDP per capita and population density, and surprisingly, neither of them is statistically significant. To be specific, the coefficients on per-capita GDP are positive but they shrink in magnitude when other factors are valued. In other words, the attractiveness of a city nowadays no longer merely depends on its economic condition (Cheng & Yang, 1998; You et al., 2018). Unlike per-capita GDP, population density has exerted negative effects on “pushing” emigrants and “pulling” immigrants as densely populated regions usually have problems of traffic congestion, indecent settlement and social inequalities (Neirotti, De Marco, Cagliano, Mangano, & Scorrano, 2014; Tan, Xu, & Zhang, 2016).

In contrast to the ambiguous role of demographic and macro-economic factors, the impact of industry development is quite clear and statistically significant at conventional levels. Generally, an increase in the share of high-tech manufacturing industries encourages a greater number of migrations, showing that cities are more ambitious in high-tech industry development; for example, Wuhu, Anhui Province and Wuxi, Jiangsu Province, indeed appeal to a large number of talents, which conformed to our second hypothesis. A rapidly growing

Table 2
Estimating results of specification regressing talent in- and out-flows.

	Inflows				Outflows			
	Model (1)	Model (2)	Model (3)	Model (4)	Model (1)	Model (2)	Model (3)	Model (4)
$\ln Dep_i$	0.96*** (0.16)	0.69 (0.49)	0.74** (0.35)	0.91 (0.43)	0.85*** (0.14)	0.69*** (0.27)	0.78*** (0.27)	0.38 (0.40)
$\ln pcgdp_i$	2.41 (1.98)	3.66 (2.59)	3.49 (2.41)	8.12 (4.04)	−2.49 (1.76)	−2.80 (1.98)	−2.49 (1.62)	−5.49 (3.53)
$\ln popden_i$	−0.75 (1.34)	−1.02 (2.66)	−1.03 (2.54)	−2.26 (3.13)	1.45 (1.00)	1.33 (0.86)	1.28 (0.82)	2.27* (1.22)
$htshare_i$	5.26** (2.38)	7.68** (4.00)	5.75 (4.08)	3.96* (2.19)	−3.72*** (1.95)	−4.30** (2.13)	−3.64** (1.83)	−4.43** (2.25)
$talreghigh_i$		0.42** (0.80)		0.42 (0.82)		−0.33** (0.37)		−0.35 (0.56)
$talreglow_i$			0.45** (0.68)	1.06 (1.01)			−0.25* (0.60)	−0.28 (0.76)
<i>Amenities</i>								
$hospnum_i$								
$univnum_i$								
$ictcov_i$								
$greens_i$								
Number of instruments	8	9	9	12	8	9	9	12
SARGAN <i>p</i> -value	0.3072	0.1997	0.1599	0.4101	0.3051	0.3329	0.3047	0.3851
AR(2)	0.6253	0.4624	0.3946	0.3283	0.6334	0.3900	0.3193	0.3793

	Inflows				Outflows			
	Model (5)	Model (6)	Model (7)	Model (8)	Model (5)	Model (6)	Model (7)	Model (8)
$\ln Dep_i$	0.62*** (0.13)	−0.22 (0.53)	−0.02 (0.57)	−0.25 (0.61)	0.63*** (0.12)	0.01 (0.32)	−0.12 (0.70)	−0.09 (0.52)
$\ln pcgdp_i$	0.67 (2.04)	1.58 (3.03)	−0.17 (2.34)	1.61 (4.16)	−3.65 (1.89)	−1.51 (2.25)	−0.54 (2.55)	−0.82 (2.86)
$\ln popden_i$	−0.86 (1.60)	7.57 (9.62)	11.01 (13.73)	8.14 (9.59)	1.42 (1.43)	−1.32 (4.34)	−4.90 (11.89)	−6.45 (8.05)
$htshare_i$	6.34*** (2.09)	7.46 (5.40)	6.08 (4.83)	7.32 (5.35)	−1.19** (1.40)	−8.72* (5.05)	−5.84 (5.95)	−3.67 (10.00)
$talreghigh_i$		2.38* (1.29)		2.46 (3.18)		−1.66** (0.85)		−0.48 (2.24)
$talreglow_i$			2.51* (1.45)	0.02 (3.03)			−2.75* (2.51)	−2.16 (3.51)
<i>Amenities</i>								
$hospnum_i$	−0.01 (0.02)	0.03 (0.03)	0.03 (0.03)	0.03 (0.03)	0.00 (0.02)	−0.01 (0.03)	−0.03 (0.03)	−0.03 (0.04)
$univnum_i$	0.10 (0.09)	0.16*** (0.05)	0.13*** (0.05)	0.16** (0.07)	−0.11*** (0.04)	−0.14 (0.05)	−0.12** (0.05)	−0.13** (0.06)
$ictcov_i$	6.35*** (1.66)	2.58 (3.63)	2.18 (2.13)	2.46 (3.27)	−3.83** (1.85)	−2.91 (3.51)	−0.78 (3.79)	−0.62 (4.16)
$greens_i$	−4.79 (8.52)	−7.35* (8.87)	−8.99 (9.81)	−13.03* (12.42)	1.33 (1.72)	6.07 (8.62)	7.93 (8.82)	8.47 (14.76)
Number of instruments	12	13	13	19	12	13	13	19
SARGAN <i>p</i> -value	0.1659	0.4126	0.4251	0.3007	0.1656	0.3889	0.3810	0.1815
AR(2)	0.6000	0.2261	0.1326	0.3261	0.4888	0.2679	0.1758	0.3578

Note: *, **, *** denote statistically significant at 10%, 5% and 1% level. Robust standard errors are in parentheses. City and year dummies are included in all models. The lagged dependent variables, GDP per capita, talent incentives for higher human capital and lower human capital are treated as endogenous variables; Second- and higher-order lags of the levels with a maximum of lag length of 3 are used to instrument endogenous variables. No external instrumental variables are used.

momentum in the industrial scale may imply more job vacancies, the importance of which in talent attraction has been evidenced in a rich body of empirical studies (Tassey, 2014; Wu, Wei, Li, & Yuan, 2018).

Turning to policy variables, the coefficients on talent incentives, be it high-skilled or low-skilled talent, are positive and statistically significant at conventional levels when separately estimated as reflected by Models (2) and (3), but turns non-significant in model (4) when they are combined. This almost indicates the mutual exclusiveness of the two regimes. Furthermore, only slight differences occur when all amenities variables are taken into consideration even though the imposition does

not alter any of the results with regard to the significance of talent policies. It can be seen that talent incentives of lower intensity outperform those of higher intensity in talent attraction, as an increase in the former associated with an increase in talent inflows, on average, is larger than the latter. This is almost certainly driven by the fact that few applicants are eligible to conform to the more rigorous requirements of immigration selection in commensurate with the richer rewards under the higher-intensity mechanism. For example, applicants are often required to have a doctoral degree, extensive management experience or hold an occupation in urgent demand. In this case, it is not difficult to

understand why the implementation of policies with lower intensity plays a more decisive role in housing and sustaining greater numbers of manufacturing talents.

In addition to the regulatory effects, the importance of urban amenities has also been emphasized, notably for high-skilled migrants (Chen & Rosenthal, 2008; Greenwood & Hunt, 1989). However, our empirical estimates seem to show a diverging effect. Only the provision of higher education, proxied by the number of universities, exhibits a robust positive and statistically significant effect in talent attraction and retention. No evidence is found that the provision of health services and Internet access are decisive for migratory decisions due to their lowest coefficients and consistent insignificance. The green space area proxying environmental quality, surprisingly, even operates as a drawback for talent attraction. It seems that the quality-of-life related factors, in the period we have studied, play little role in destination choices within the YRDU. Similar findings have been documented by many previous studies exploring the role of urban amenities (Dorfman, Partridge, & Galloway, 2011; Mattoni, Gugliermetti, & Bisegna, 2015).

6. Discussions

The traditional gravity model is being augmented by an increasing number of empirical evidence in high-skilled migration decisions in many countries (Kerr, Kerr, Özden, & Parsons, 2017). The migration patterns of highly skilled workers vary greatly across countries. For instance, Becker and Teney (2020) concluded three classes of migration patterns, namely traditional labor migrants, career seekers, and family settlers. They also found that talents in Western Europe are more likely to belong to the family settlers' class, while those in Eastern Europe are more of the traditional class who tend to move due to economic and work-related reasons. However, the drivers for high-skilled migration decisions are similar across countries. Evidence from several Organization for Economic Co-operation and Development (OECD) countries reveals that labor market conditions or socio-economic amenities comparably have more significant and positive impacts on encouraging higher flows of high-skilled workers than cultural factors, e.g., language and social relations (Cleave & Arku, 2020; Kerr et al., 2017). In South Korea, researchers are more responsive to wage dispersion than job opportunities (Lee & Kim, 2018). In Canada, immigrants view the economy and housing as the key drivers of the evaluation of cities (Cleave & Arku, 2020).

As cities today compete and cooperate, the local authorities usually brand cities by imaging themselves for some narrowly-defined targets, e.g., investors, firms, and individuals, in the context of urban planning and economic development goals (Belabas, Eshuis, & Scholten, 2020; Zenker & Jacobsen, 2015). For example, city branding often serves the purpose of stimulating economic growth in the U.S., while it is utilized to improve various aspects of socio-economic well-being in Europe (Anttiroiko, 2015). Therefore, city branding is more specifically reflected in regional development plans than broadly advertising across the media (Prilenska, 2012).

Notably, economic city branding may be used in the YRDU with particular reference to the industrial specialization or the upgrading of traditional labor-intensive manufacturing industries (Björner, 2014). To achieve this goal, the role of education resources in building high-tech cities is crucial. High-quality universities annually produce a considerable number of well-educated students who are also would-be residents of where those universities are located (Anttiroiko, 2015). For example, Stanford University and UC Berkeley have substantially contributed to the formation and the rise of the Silicon Valley (Anttiroiko, 2015). Such perspectives help to re-profile the macro-area in terms of knowledge and innovation, thereby providing the right way of

communicating city branding in the context of local industrial development policy (De Jong et al., 2018). Furthermore, economic city branding can be built with a broader range of social well-being promotions, such as logistics, transportation, tourism, and the ecosystem, as high-skilled workers strongly value regional amenities in their migration decisions (Björner, 2014; De Jong et al., 2018).

In 2010, China positioned the whole YRDU as a world-class center for high-end manufacturing and modern services industries. The local governments need to profile cities with various factors that matter for the target audience and ensure that their branding strategies have been well communicated both internally and internationally.

7. Conclusion

As the endogenous growth theory suggests that investments in human capital and technological innovation mutually contribute to economic growth, the attraction of talents is expected to bring local, provincial, and regional benefits. As the most developed urban cluster in China, the YRDU has an average share of 40% of advanced manufacturing industries in 2018, twice as that in 2008, while the migration scale of high-tech talents in 2018 is 13 times larger than that in 2015. Particularly, Shanghai is still the largest migrants sending and receiving municipality in the region, followed by Hangzhou, Nanjing and Suzhou. The YRDU has recently implemented a series of talent incentives aimed at attracting and selecting high-skilled workers to meet their desired goals of high-tech manufacturing industry development and economic growth.

Using a panel of 26 cities in the YRDU from 2008 to 2018, this study presents the first judicious assessment of the effectiveness of the latest talent policies on high-skilled workers, internalizing the relationship between high-tech sectors and economic growth. The objective is to empirically study the macroeconomic determinants of high-tech migration in this region. The annual unilateral flows of workers who expect to get a job in the high-tech manufacturing sectors in the YRDU are estimated with the unique dataset provided by Boss Zhipin, one of the largest online recruiting platforms in China. To take into consideration the potential endogeneity and the dynamic nature in the relationship examined, this study uses a two-equation panel model estimated by the system generalized method of moments (sys-GMM) estimator.

As empirically tested, internal support from local government, universities and financial institutions on the fast development of high-tech sectors are seemingly eclipsed by foreign investments, which is presumably induced by the policy inertia towards traditional industries. Even more surprisingly, local government expenditure on science and technology is found to impose a negative impact on the high-tech sector performance as a consequence of crowding-out effects and the lacking of effective auditing mechanism. Nevertheless, the economic significance of all these variables is worth noting in stimulating high-tech manufacturing industries.

We have proposed three hypotheses regarding three conditions under which high-tech talents are prone to regions with such characteristics. For our first hypothesis, we find a countervailing effect in economic and demographic conditions. Although high-tech talents tend to leave for regions with a higher stage of economic development, the ruling role of economic attractiveness in their destination choices is gradually depressed considering that their migration decisions are not solely economic-driven. Talents are more likely to leave densely-populated places when problems like traffic congestion, indecent settlement and social inequalities overwhelm them.

Our second hypothesis is accepted and empirical findings prove that cities that are more ambitious in developing high-tech industries indeed

appeal to larger numbers of qualified immigrants, mainly due to the fact that a rapidly growing momentum in the industrial scale often indicates more jobs and possibly higher wage returns.

Our third hypothesis regarding the efficacy of talent incentives is also accepted. Two different policy tools with higher and lower intensities in rewards are examined both in isolation and in combination. We find that both policies are positive in the attraction and retention of the high-tech cohort, but the low-intensity incentive is more decisive than the higher one as the latter has more rigorous criteria of talent selection.

Even though the quality-of-life factors, i.e., the standard of living, the quality of schooling, the provision of health services, and the effectiveness of institutions, have recently drawn an increasing attention to researchers, amenities are found to be playing little role in destination choices in this study.

Seen from the other side, the YRDUA has exhibited a shifting landscape of regional collaboration from the Shanghai-centric pattern to a more balanced network spatial pattern in which population concentration and dispersion were combined. Our research adds to the literature by assessing the efficacy of recent talent competition and exploring the migration dynamics in the high-tech manufacturing

pipeline within the YRDUA by aggregating a micro database in the context of macro analysis, and it is our belief that our models can be applied to the study of similar mega-city agglomerations. With the help of our unique dataset, some other aspects of high-skilled migration, notably the consequences of the “brain drain and return” on the origin cities, can be further studied in future work.

CRediT authorship contribution statement

Lu Wang: Conceptualization, Methodology, Visualization, Writing - Original draft preparation.

Yanbo Xue: Supervision, Project administration, Investigation, Writing - Review and Editing.

Meng Chang: Resources, Data Curation, Writing - Review and Editing.

Chen Xie: Software, Visualization, Modeling, Data Curation.

Declaration of competing interest

None.

Appendix A

Table A1
Migration efficiency of 26 cities in 2015–2018.

	2015	2016	2017	2018
Anqing	0.034	−0.086	−0.022	−0.009
Changzhou	0.129	−0.005	−0.002	0.046
Chizhou	0.333	−0.176	−0.118	0.105
Chuzhou	0.357	0.063	−0.048	0.018
Hangzhou	−0.019	0.068	0.017	−0.009
Hefei	0.016	−0.081	0.020	0.045
Huzhou	0.102	−0.136	0.016	0.027
Jiaxing	−0.039	−0.113	−0.093	0.026
Jinhua	0.059	−0.088	−0.007	0.128
Ma'anshan	0.440	−0.280	−0.108	−0.077
Nanjing	−0.004	0.023	0.023	−0.011
Nantong	0.165	−0.057	−0.002	0.047
Ningbo	0.016	−0.045	−0.026	0.036
Shanghai	−0.081	0.037	0.020	−0.043
Shaoxing	−0.200	−0.250	0.055	0.021
Suzhou	0.011	−0.170	−0.045	−0.006
Taizhou	0.034	0.010	−0.076	0.084
Taizhou	−0.174	−0.169	−0.077	0.047
Tongling	0.059	0.111	−0.313	0.076
Wuxi	−0.116	−0.054	−0.011	0.031
Wuhu	−0.200	−0.093	−0.058	0.024
Xuancheng	−0.176	−0.091	0.084	0.100
Yancheng	−0.227	−0.237	−0.167	0.112
Yangzhou	−0.038	−0.022	−0.046	−0.022
Zhenjiang	0.130	−0.138	−0.075	0.023
Zhoushan	0.000	−0.455	−0.130	0.104

Table A2

Summary of Talent Incentives of YDRUA. Data Source: authors' collection; raw data are assembled from governments official websites.

		2017		2018	
		higher-intensive incentives	lower-intensive incentives	higher-intensive incentives	lower-intensive incentives
Shanghai	Shanghai				
Southern Jiangsu	Nanjing				
	Wuxi				
	Changzhou				
	Suzhou				
	Yangzhou				
	Zhenjiang				
	Yancheng				
	Nantong				
	Taizhou				
Northern Zhejiang	Shaoxing				
	Taizhou				
	Hangzhou				
	Zhoushan				
	Ningbo				
	Jiaxing				
	Huzhou				
	Jinhua				
Southeast Anhui	Chuzhou				
	Wuhu				
	Anqing				
	Hefei				
	Ma'anshan				
	Xuancheng				
	Chizhou				
	Tongling				

In the “war of talents” in China, cities typically emphasize two categories of demand, namely for high-end talents and for college graduates with technical skills. The high-end talent program principally targets innovative technical personnel or outstanding management personnel recognized by core national, provincial or municipal talent programs, or urgently-needed talent across a range of sectors including scientific research and development teams of key national, provincial or municipal innovation platforms.

As the YDRUA envisages a blueprint of advanced manufacturing, there has been a surge of investment in the local talent pool. Local municipal governments promise to provide various subsidies and policy incentives for eligible talents who would come to work in the respective city, ranging from housing subsidies to relaxed requirements on household registration. Specifically, the policy incentives for high-end talents also include income tax break, startup funding and quality education and health services. We summarized and compared different policies within the YDRUA, and found that incentives for eligible high-end talents are much higher with a maximum of 8 million yuan (around \$1.13 million) in total, which is categorized as “high-intensity policy” in our study. Subsidies for college graduates range from 10,000 to 200,000 yuan for renting welfare or one-off living subsidies, which are categorized as “low-intensity policy”. Detailed information of the incentives can be accessed from the official websites of different local government units.

Table A3

Description of variables.

Variable	Obs.	Mean	Std. dev.	Min	Max
High-tech Share (%)	286	0.35	0.12	0.01	0.62
GDP per capita (1000 yuan in 2005 price)	286	72.51	34.15	12.60	174.30
Share of government expenditure on science and technology (%)	286	0.04	0.02	0.01	0.13
Inward foreign direct investments (million yuan)	286	2.31	3.15	0.48	111.51
University numbers	286	13.02	16.38	1.00	68.00
Financial resources (billion yuan)	286	843.58	1511.02	22.64	11246.17
Migration net flows	104	1011.67	2962.57	4.00	21144.00
Population density (number of people per square kilometer)	104	828.35	679.64	171.01	3822.39

(continued on next page)

Table A3 (continued)

Variable	Obs.	Mean	Std. dev.	Min	Max
Hospital numbers	104	93.77	88.14	0.00	349
ICT coverage (%)	104	0.3	0.13	0.07	0.85
Share of green space (%)	104	13.35	16.50	1.00	67
Incentives with higher-intensity rewards	104	0.23	0.42	0.00	1.00
Incentives with lower-intensity rewards	104	0.21	0.41	0.00	1.00

To obtain a first impression of all variables, Table A3 describes the basic characteristics of data used for the construction of the two panels. Wuxi, Jiangsu Province has the highest level of GDP per capita, while the lowest is found in Anqing, Anhui Province. In terms of gross output value, high-tech manufacturing industries take 40% of the above-scale industries on average across the YRDU, and there are wide gaps across the 26 cities. Hefei, Anhui Province has the largest share of advanced manufacturing (62%) in 2017 and 2018, while Anqing has only 1% in 2008 but has seen an annual growth of 30%. The total migration flows in and out the region in 2018 are 13 times larger than four years ago. Shanghai, undoubtedly, is both the largest sender and receiver of high-tech talents, followed by Hangzhou, Nanjing and Suzhou. Similar trends can also be observed in other macro-economic factors.

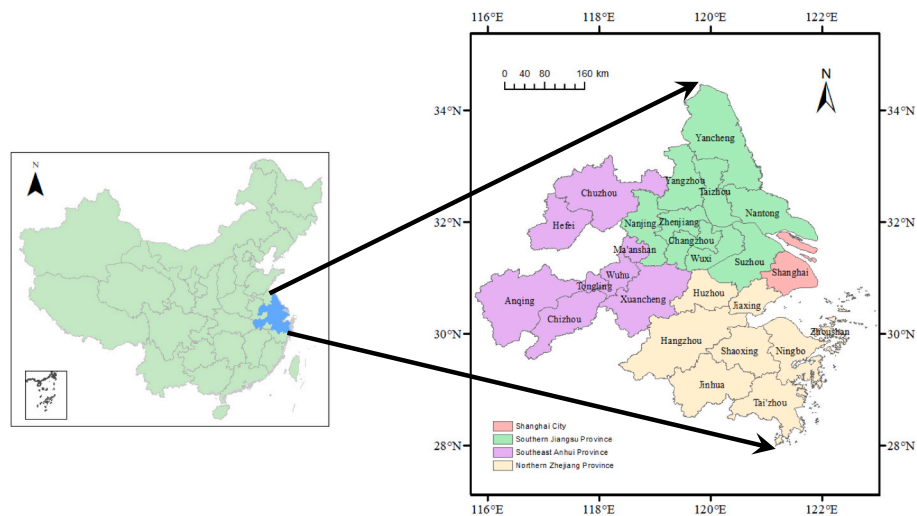


Fig. A1. Yangtze River Delta Urban Agglomeration. The Yangtze River Delta (YRD) has seen a dramatic increase in its population, economic vitality and urbanization progress since the implementation of Openness and Reform Policy in 1978. In the following years, YRD has experienced several changes in its development practices until the latest confirmation in 2016 of its membership with more cities from Anhui joining the club. It includes one municipality and 25 prefecture-level cities that cover four jurisdiction units, i.e., Shanghai City, Northern Zhejiang Province, Southern Jiangsu Province, and Southeast Anhui Province, which is officially named as the Yangtze River Delta Urban Agglomeration (YRDU), which enjoys a geographic advantage as the intersection zone of the “Belt and Road Initiative” and the “Yangtze River Economic Belt”.

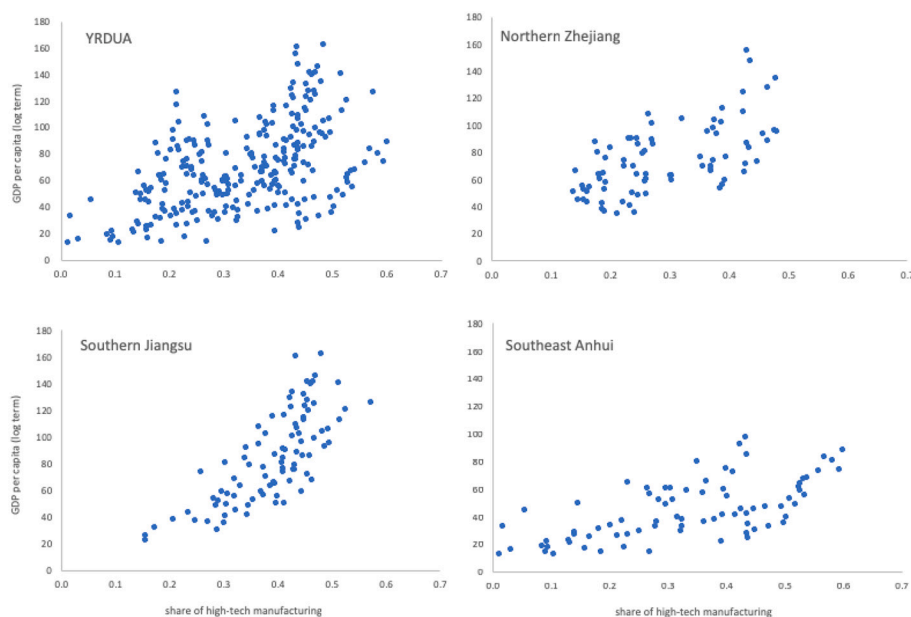


Fig. A2. Relationship between GDP per capita and share of high-tech manufacturing industries (pooled 2008-2018). (Source: Statistical Yearbooks, authors' calculation).

Fig. A2 illustrates the relationship between economic growth and the scale of the high-tech manufacturing industry in the YRDU and its three sub-regions respectively. As we can see from the macro area, the development of advanced manufacturing industries is positively correlated with economic growth, and similar trends appear in northern Zhejiang and southern Jiangsu. A special case of southeast Anhui shows a more rapid advance in high-tech manufacturing industries than the commensurate pace of economic growth.

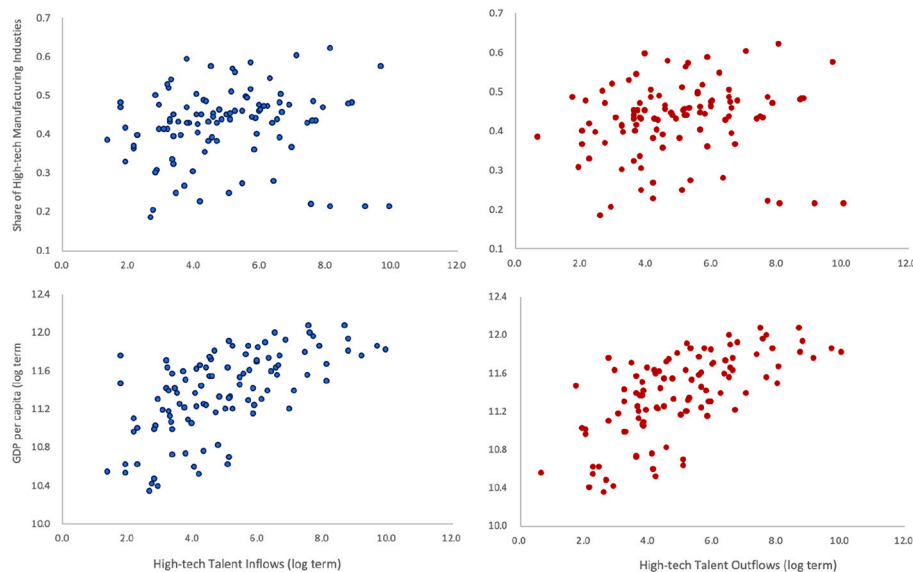


Fig. A3. Relationship between GDP per capita, high-tech manufacturing industries and regional high-tech inflows and outflows (pooled 2015–2018). (Source: Statistical Yearbooks, Boss Zhipin).

In Fig. A3, the two main examined factors, i.e., GDP per capita and the share of high-tech manufacturing industries, are seen positively associated with both in- and out-migration flows. Specifically, the trend in the industrial effects seems more explicit.

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